References and Source Map

Scientific Communication: Writing, Slide Design and Presentation Skills

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How these sources were checked

Every source below was opened and read on 4 November 2026, in full or, where stated, as an abstract or bibliographic record only. Some publisher pages refuse automated access, so a few sources were read through PubMed Central, Europe PMC, an author's copy or a university repository. In those cases the link given is the most stable one for readers. Long pages were sometimes read with text extraction or a summarising tool, so every figure quoted in the course was then checked by eye against the source text. Links were tested by script, and the results are in materials/references/link_check.csv.

Evidence types

Label	Meaning	How to use it
Research finding	An empirical study or systematic review.	State the design, the setting and the limits. Learning or online studies do not automatically transfer to research talks.
Review or consensus	A narrative or methods review, meta-analysis, or expert statement.	Reliable on principles. Check how far it applies.
Professional guidance	Advice from experienced people or organisations.	Useful, but not evidence of effect. Most slide advice is of this kind.
Standard or policy	A standard, or a policy of an institution or journal group.	Binding only where it applies, so check yours.
Teaching resource	Teaching material from a university or a method's authors.	Practical. Often written by advocates.
Teaching recommendation	Our own device, such as the four moves or the rubric.	Labelled as ours in the materials. Not a finding.

Rules of thumb in this course

We searched for experiments that set a slides-per-minute rate, a six-by-six word limit, or a minimum point size for projected slides, and found none. This was a targeted search, so the materials say that we found no evidence, and they do not say that none exists.

Rule	What the sources say	Practical limit
One slide per minute	Bourne 2007 calls more than one visual per minute too many, as a personal rule of thumb. Naegle 2021 advises a minute or less per slide. Garner & Alley 2013 tested a talk at about 1.4 slides per minute, although rate was not the variable tested.	A planning check only. Density and build matter more than count.
Six words, six lines	No primary source found. The closest statement is a summary of World Blind Union guidance, which gives no evidence for its numbers. Less text alongside speech has reasoned and indirect support.	Decide what must be read, and move the rest.
Minimum 24 or 30 point	Published advice ranges from 18 pt (Microsoft) through 18 to 24 pt body text (Penn State) to 32 pt (WBU summary). W3C gives no number. Legge & Bigelow 2011 show that legibility depends on angular size. WCAG's 18 pt defines large text for the contrast rule and is not a font minimum.	Test from the back of the actual room, or at reduced size.

Corrections we made to common claims

- Garner et al. 2009 is a slide audit. The learning experiment is Garner & Alley 2013, which tested a bundle of changes on one recorded talk.
- Mayer & Moreno's redundancy effect concerns identical on-screen text with narration and animation. With no animation, narration plus text did better than narration alone. It does not show that text on slides is harmful in general.
- Akl et al. 2011 found relative figures looked larger and persuaded more than absolute ones, with no difference in understanding. Its natural-frequency finding applies to diagnostic and screening information.
- Gopen & Swan 1990 is an argued essay with no reader testing, and the authors say their principles are not rules.
- HHS de-identification guidance gives no rule on small cells. We use the Public Health Scotland protocol as one worked example and tell readers to find their own rule.
- The CDC Clear Communication Index score is a percentage of applicable items. CDC gives no validation for its 90 cut-off, and we call it a convention.
- The former plainlanguage.gov guidelines page now redirects to digital.gov. The Guidelines are reposted by the Center for Plain Language.

Source-to-content map

These tables show which sources support each module, and what each source is used for. M1 to M6 are the course modules. F covers facilitation and feedback design.

M1: Audience, purpose and message.

Source	Type	What it supports here
Bourne 2007	Professional guidance	Know the audience, rehearse with colleagues, leave room for questions. More than one visual per minute is called too many, as a personal rule of thumb.
CDC Index	Professional guidance	Before scoring a public material, state its audience, objective and main message. The index scores applicable items as a percentage, with 90 as CDC's passing level.
Plain Language	Professional guidance	Start with the audience, use active voice and real verbs where suitable, and test with readers.
Digital.gov PL	Professional guidance	Plain language matches words and detail to what readers know. It does not mean oversimplifying.

M2: Scientific writing.

Source	Type	What it supports here
Gopen & Swan 1990	Teaching resource	Readers look to the start of a sentence for context and to the end for emphasis. Keep the verb close to its subject.
Plain Language	Professional guidance	Start with the audience, use active voice and real verbs where suitable, and test with readers.
Digital.gov PL	Professional guidance	Plain language matches words and detail to what readers know. It does not mean oversimplifying.
MIT abstract	Teaching resource	Write the abstract last, and let the reader decide whether the paper is worth reading.
UNC abstracts	Teaching resource	Informative abstracts stand in for the work and give main results and conclusions.

Source	Type	What it supports here
Manchester Phrasebank	Teaching resource	Writers soften claims where uncertainty remains, most often when explaining findings.
Altman & Bland 1995	Review or consensus	A non-significant result usually shows a lack of evidence of a difference, not evidence of no difference.
ASA 2016	Review or consensus	P-values do not measure effect size or importance, and decisions should not rest on a threshold alone.
Greenland 2016	Review or consensus	P above 0.05 does not show no effect. Interval width indicates precision.
SAMPL	Professional guidance	Give numerators and denominators for percentages, and effect estimates with precision, usually a 95% CI.
Sumner 2014	Research finding	About a third of 462 university press releases made stronger causal claims than the journal article, and exaggeration in news was associated with exaggeration in releases.
Haber 2018	Research finding	A third of the academic papers reviewed used language too strong for their inference.
Haber 2022	Research finding	Many reviewers read some causal meaning into the word association, and recommendations often imply more causality than the linking sentence.
Hernan 2018	Review or consensus	State a causal aim openly rather than hiding it behind associational wording.
van der Bles 2019	Review or consensus	Separates uncertainty about a number from uncertainty about the quality of the evidence behind it.
van der Bles 2020	Research finding	In five experiments, numeric ranges cost little trust, and verbal uncertainty cost more.
GRADE 26	Professional guidance	Convey both effect size and certainty. Wording steps down from 'results in' to 'likely' to 'may', and vague phrases such as 'limited evidence' should be replaced.
Gigerenzer 2007	Review or consensus	Recommends absolute risks and frequency statements over relative risks and single-event probabilities.
Akl 2011	Research finding	Relative reductions looked larger and persuaded more than absolute ones, with no difference in understanding. So do not give the relative figure alone.
STROBE	Standard or policy	Abstracts should give key results as numbers with effect estimates and uncertainty. Adjustment does not by itself establish causation.
CONSORT Abstracts	Standard or policy	Readers often judge a trial from its abstract alone, so report results for each group with effect size and precision.

M3: Presentation structure.

Source	Type	What it supports here
Bourne 2007	Professional guidance	Know the audience, rehearse with colleagues, leave room for questions. More than one visual per minute is called too many, as a personal rule of thumb.
Naegle 2021	Professional guidance	One idea per slide, design for a distracted listener, keep backup slides, prepare for technical failure.
AE defence	Teaching resource	Present a slice of the work slowly enough to follow and show where it fits the field.
Imperial viva	Standard or policy	A private oral examination with no prescribed length or format, where the student explains and defends the thesis.

Source	Type	What it supports here
Minnesota ECE	Standard or policy	A public presentation followed by a closed session with the committee.
UCLA Neuroscience	Standard or policy	A public talk of 45 to 60 minutes accessible to a general audience, open questions, then a closed session.
Halmstad	Teaching resource	A public defence with an external opponent, usually lasting two to three hours, and a board deciding in private.
MIT BE defence	Teaching resource	Build background for the whole room, organise for clarity rather than chronology, and close with significance, limits and next steps.

M4: Slide design and data communication.

Source	Type	What it supports here
Bourne 2007	Professional guidance	Know the audience, rehearse with colleagues, leave room for questions. More than one visual per minute is called too many, as a personal rule of thumb.
Naegle 2021	Professional guidance	One idea per slide, design for a distracted listener, keep backup slides, prepare for technical failure.
Alley, AE site	Teaching resource	A complete-sentence headline supported by visual evidence, with secondary detail in the notes.
AE tutorial	Teaching resource	Steps for building a headline and evidence, then explaining the visual in your own sentences.
Penn State set	Teaching resource	Suggests 28 pt headlines, 18 to 24 pt body text, and practice of about five times the talk length, as a rule of thumb.
Garner 2009	Research finding	Content analysis of about 2,000 slides: phrase headlines and bullets were the common pattern, including among PhD students presenting in English as an additional language.
Garner & Alley 2013	Research finding	Randomised experiment with 110 engineering undergraduates: the assertion-evidence version of a recorded talk gave better comprehension and recall, with less reported effort.
Farkas 2010	Professional guidance	Critique: comparing against poorly designed slides is a weak test, and competent bullets do no harm.
Mayer & Moreno 2003	Review or consensus	Removing extraneous material helped learning. Identical on-screen text harmed learning alongside animation, but with no animation, narration plus text did better than narration alone.
Kosslyn 2012	Research finding	Audiences most often reported too much material before the next slide, and observers were poor at spotting flaws.
Rougier 2014	Professional guidance	Decide the message first, adapt figures for the medium, do not trust defaults, avoid distortion and decoration.
Heer & Bostock 2010	Research finding	Replicated classic perception results: judgements of position beat length, and length beat angle and area.
Cleveland & McGill 1984	Research finding	Original work on perceptual ordering, cited as the study Heer and Bostock replicated.
Weissgerber 2015	Research finding	Different datasets can give the same bar chart, so show individual points for small continuous datasets.
Correll 2020	Research finding	Truncated axes made changes look more severe, even when readers could read values, but the authors reject a blanket start-at-zero rule.
Pandey 2015	Research finding	In online experiments truncated bars made a difference seem larger, and an inverted axis reversed the message for most readers.

Source	Type	What it supports here
Wilke 2019	Teaching resource	Bars must start at zero, dot plots need not. Use few categorical colours, label directly, test in a colour-vision simulator. Gives the Okabe and Ito hex values.
Crameri 2020	Review or consensus	Rainbow colour maps can distort data. Perceptually ordered maps stay readable in greyscale.
Okabe & Ito	Professional guidance	Add shape, position, line type or direct labels so that information survives without colour. Do not point things out by colour name alone.
WCAG 1.4.3	Standard or policy	Level AA asks for 4.5:1 for text and 3:1 for large text.
WCAG 1.4.1	Standard or policy	Colour must not be the only way of conveying information.
WCAG2ICT	Standard or policy	Lists presentations among non-web documents and treats the contrast criterion as applicable to documents.
W3C Events	Professional guidance	Make text readable from the back of the room, limit text, speak clearly, cover what is on the slide, repeat questions.
Microsoft	Professional guidance	Give every slide a unique title, check reading order, write alt text, and use 18 pt or larger.
WBU summary	Professional guidance	Suggests no less than 32 pt, and short lines, for audiences with print impairments.
Legge & Bigelow 2011	Review or consensus	Legibility depends on angular size at the eye, so a point size means little without screen size and distance.

M5: Delivery, rehearsal and questions.

Source	Type	What it supports here
Bourne 2007	Professional guidance	Know the audience, rehearse with colleagues, leave room for questions. More than one visual per minute is called too many, as a personal rule of thumb.
Naegle 2021	Professional guidance	One idea per slide, design for a distracted listener, keep backup slides, prepare for technical failure.
Penn State set	Teaching resource	Suggests 28 pt headlines, 18 to 24 pt body text, and practice of about five times the talk length, as a rule of thumb.
Okabe & Ito	Professional guidance	Add shape, position, line type or direct labels so that information survives without colour. Do not point things out by colour name alone.
W3C Events	Professional guidance	Make text readable from the back of the room, limit text, speak clearly, cover what is on the slide, repeat questions.
Munro & Derwing 1995	Research finding	A strong accent did not necessarily reduce how well speech was understood.
Derwing & Munro 2009	Review or consensus	Pronunciation work has moved from native-like speech towards easy intelligibility.
Ebrahimi 2019	Review or consensus	Structured psychological interventions reduced fear of public speaking in 30 randomised trials.
Streeter & Miller 2011	Professional guidance	Rehearse questions, repeat them, say when you do not know, and offer to follow up. Do not argue with hostile questions.
Dunphy 2007	Teaching resource	A four-step answer for speakers of English as an additional language: acknowledge, check, answer, check again.
Waterloo	Teaching resource	Do not bluff. Admitting you do not know costs less credibility than a wrong answer.
NIU	Teaching resource	Separate valid criticism from attack and answer the valid part.

Source	Type	What it supports here
Hattie & Timperley 2007	Review or consensus	Feedback about the task and how to improve it helps most. Feedback aimed at the person rarely helps.
van der Bles 2019	Review or consensus	Separates uncertainty about a number from uncertainty about the quality of the evidence behind it.
van der Bles 2020	Research finding	In five experiments, numeric ranges cost little trust, and verbal uncertainty cost more.

M6: Accuracy and responsible use.

Source	Type	What it supports here
CDC Index	Professional guidance	Before scoring a public material, state its audience, objective and main message. The index scores applicable items as a percentage, with 90 as CDC's passing level.
Sumner 2014	Research finding	About a third of 462 university press releases made stronger causal claims than the journal article, and exaggeration in news was associated with exaggeration in releases.
ICMJE	Standard or policy	Sets four authorship criteria, requires permission for acknowledgement, protects patient identity, and says humans remain responsible for AI-assisted work, which must be disclosed.
COPE AI	Standard or policy	AI tools cannot be authors. Authors are responsible for all content and must disclose use.
HHS de-identification	Standard or policy	Initials and parts of identifiers still count as identifiers, and de-identification does not bring risk to zero.
GDPR Recital 26	Standard or policy	Pseudonymised data remain personal data. Identifiability depends on means reasonably likely to be used.
PHS disclosure	Professional guidance	Small cells, and differences between tables, can identify people. Rates can reveal small counts, and small numbers can expose a clinician.
Walters & Wilder 2023	Research finding	Many citations from two 2023 models pointed to works that do not exist, so a plausible citation is not a verified one.
Peters & Chin-Yee 2025	Research finding	Most models tested broadened claims when summarising science, and asking for accuracy did not help.

F: Facilitation and feedback design.

Source	Type	What it supports here
Munro & Derwing 1995	Research finding	A strong accent did not necessarily reduce how well speech was understood.
Derwing & Munro 2009	Review or consensus	Pronunciation work has moved from native-like speech towards easy intelligibility.
Hattie & Timperley 2007	Review or consensus	Feedback about the task and how to improve it helps most. Feedback aimed at the person rarely helps.
Double 2020	Review or consensus	Peer assessment improved academic performance on average, with a smaller estimate in tertiary students.

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Limits of each source

Source	Limit to keep in mind
Bourne 2007	Opinion from experience. No data behind any rule.
Naegle 2021	The one-minute and six-element rules are advice, not tested claims.
Alley, AE site	Written by the method's advocate.
AE tutorial	Advocacy source. No numeric limits given.
AE defence	No slide count or timing given.
Penn State set	Sizes and the practice rule carry no evidence.
Garner 2009	Descriptive, no audience outcome. Not a random sample.
Garner & Alley 2013	A bundle of changes on one recorded talk, one university. An author is the method's advocate.
Farkas 2010	Informal memo, not peer reviewed. Predates the 2013 experiment.
Mayer & Moreno 2003	Short computer lessons with students, not live talks.
Kosslyn 2012	Prevalence and annoyance, not learning. Convenience samples.
Rougier 2014	Editorial advice.
Heer & Bostock 2010	Proportion judgements online with crowd workers, not projected slides.
Cleveland & McGill 1984	Full text not opened. We rest teaching claims on Heer and Bostock.
Weissgerber 2015	One field. Applies to continuous data, not to counts or proportions.
Correll 2020	Small online samples, context-free charts.
Pandey 2015	Paid online participants and simple charts.

Source	Limit to keep in mind
Wilke 2019	One expert's judgement. Licence bars derivatives, so nothing is reproduced here.
Crameri 2020	Concerns continuous colour maps. Audience effects inferred.
Okabe & Ito	Expert guidance. Prevalence figures vary by population.
WCAG 1.4.3	Written for web content. The 18 pt threshold defines large text for contrast and is not a font minimum.
WCAG 1.4.1	Web scope. See WCAG2ICT for documents.
WCAG2ICT	Informative note, so slides are not bound to comply.
W3C Events	Consensus checklist, not evidence of effect.
Microsoft	Vendor documentation that changes with versions.
WBU summary	Secondary summary, with no evidence given for its numbers.
Legge & Bigelow 2011	Concerns running text at reading distance, not slides.
Munro & Derwing 1995	Abstract only. Small study, one first-language group, short utterances.
Derwing & Munro 2009	Abstract only. Narrative plenary paper.
Ebrahimi 2019	Measures fear, not talk quality. A short workshop is not such an intervention.
Streeter & Miller 2011	Experience-based advice for bioscience talks.
Dunphy 2007	One course handout. Not tested.
Waterloo	Local exam guidance. Some of its style advice differs from other sources.
NIU	General, not specific to science.
Hattie & Timperley 2007	Mostly school evidence, so applying it to adults is an extrapolation.
Double 2020	High heterogeneity. Not specific to oral presentation or to professionals.
Imperial viva	One UK university.
Minnesota ECE	One department.
UCLA Neuroscience	One programme.
Halmstad	One university. Practice varies by faculty.
MIT BE defence	Observations of defences of about 45 minutes. Does not apply to a viva.
CDC Index	Built for public health materials, not papers. A high score does not show readers understood.
Gopen & Swan 1990	An argued essay with no reader data. The authors say these are not rules.
Plain Language	Written for US government documents. The former plainlanguage.gov link now redirects.
Digital.gov PL	Shorter successor guide. Pages show no date.
MIT abstract	Tailored to basic biomedical papers. Journal rules take precedence.
UNC abstracts	Written for all disciplines. CC BY-NC-ND, so linked and not adapted.
Manchester Phrasebank	A phrase bank. Does not say how strong a claim should be.
Altman & Bland 1995	Short methods note with selected examples.
ASA 2016	Principles, not reporting rules. Does not ban p-values.
Greenland 2016	Conceptual guide. Dense for beginners.
SAMPL	Not built by formal consensus. Journal style can override details.
Sumner 2014	Correlational. One country and year. A correction exists (BMJ 2014;349:g7666), and its detail was not seen, so check it before quoting further figures.
Haber 2018	Most-shared stories only, so not representative of research at large.
Haber 2022	High-profile journals only.

Source	Limit to keep in mind
Hernan 2018	Commentary. Journals may still ask authors to avoid causal terms.
van der Bles 2019	Narrative review.
van der Bles 2020	UK public and non-health topics.
GRADE 26	Built for systematic reviews, and needs a certainty rating.
Gigerenzer 2007	Narrative review with a clear point of view.
Akl 2011	Search to 2007. Mostly hypothetical scenarios.
STROBE	A reporting guide, not a quality score. Observational studies only.
CONSORT Abstracts	Trials only. Not yet aligned with CONSORT 2025.
ICMJE	Applies to journals that follow it.
COPE AI	Differs from ICMJE on where to disclose, so follow the target journal.
HHS de-identification	US law for covered entities. Gives no small-cell rule.
GDPR Recital 26	A recital aids interpretation. Read through an unofficial copy.
PHS disclosure	One agency's protocol. Thresholds differ, so find your institution's rule.
Walters & Wilder 2023	A snapshot of old models. Shows a failure mode, not a current rate.
Peters & Chin-Yee 2025	Version-specific results that will age.

Considered and not used

Item	Why
UK ICO anonymisation guidance	The site refused automated access, so the text was not read.
Alley, The Craft of Scientific Presentations (book)	Not opened.
Crowther and Isbell 2023	Blocked. Only a search summary was seen.
Smith and Frymier 2006, on audience practice	Citation confirmed, abstract not opened.
Wong 2011, Points of view: Color blindness	Only the first paragraph was available. Palette values come from Wilke instead.
Schwabish 2014	Abstract page only, and not needed.
Vitae and University of Nottingham viva pages	No usable page was found.
Vendor pages on the 6x6 rule and 30-point claims	Seen only in search snippets.